PHAN VAN BANG

AI Engineer · Multi-Agent Systems, RAG, Voice AI

phanvanbang.id@gmail.com
+84 329 958 001 · Da Nang, Vietnam
github.com/pvbang · linkedin.com/in/pvbang
pvbang.github.io/cv

SUMMARY

AI Engineer with 3+ years building LLM products end to end (FastAPI, React, Docker, AWS): multi-agent systems, hybrid RAG and GraphRAG, voice AI, computer vision and LLM fine-tuning. Full-time at MekongAI (2023–2026), now freelancing for clients. Author of the open-source Nexus multi-agent platform: ~236K lines of first-party code, 197 MCP tool registrations across 66 modules, 1,373 test functions. Also shipped a LangGraph repair assistant with 3,300+ test functions and an AI video studio with a self-correcting quality critic.

TECHNICAL SKILLS

AI / LLM	Multi-agent orchestration, structured debate, tree-search planning, hybrid RAG (dense + BM25 + graph, RRF), GraphRAG, reranking, MCP (client, server, OAuth 2.1 PKCE), LangGraph, LangChain, LlamaIndex, CrewAI, LoRA/QLoRA (Unsloth), vLLM, structured outputs
Voice & Vision	Whisper / faster-whisper STT, TTS (OpenAI, Fish Audio, ElevenLabs, Edge), VAD, PaddleOCR, SAM3, ComfyUI / Stable Diffusion, Gemini multimodal, spaCy (vi + en), FFmpeg
Backend	Python, TypeScript, FastAPI, SQLAlchemy + Alembic, Celery, WebSocket, SSE, MQTT, OpenAI-compatible APIs, LiteLLM
Data	PostgreSQL + pgvector, Qdrant, Chroma, MongoDB, Redis, Neo4j, Firestore
Frontend	React 19, TypeScript, Vite, Tailwind CSS, Zustand, Streamlit (agent, chat and analysis UIs)
DevOps & QA	Docker Compose, AWS (EC2, S3), Nginx, RunPod (GPU serverless), Prometheus-style metrics, Pytest, Ruff, mypy, Vitest

WORK EXPERIENCE

Freelance AI Engineer

08/2026 – Present

- Build AI products for clients end to end: a LangGraph diagnostic assistant for automotive repair (hybrid RAG, SSE streaming, S3 backup and restore) and **M-Auto**, an AI YouTube video studio with a self-correcting quality critic.
- Open-sourced Nexus and published LLM tooling (mcp-prune, sentinel-scan).

AI Engineer · MekongAI

06/2023 – 07/2026

- Architected **MekongAI Nexus**, a multi-agent platform with structured debate, layered memory, hybrid RAG + GraphRAG and a continual-learning loop (details below).
- Shipped **Knowlympic** (PaddleOCR + LLM question generation, cut content-creation time for 3 English language centers) and **MetaHuman** (Whisper STT, streaming TTS, VAD, 3-tier personality memory, ESP32 voice bots over MCP and MQTT).
- Built early customer-service RAG chatbots (LangChain, LlamaIndex, Chroma) with Messenger integration, intent-routed prompt chains and per-tenant knowledge bases; later TikVault, an AI recruitment platform and an AI content generator.
- Delivered vision and GenAI services (20+ ComfyUI workflows, face-similarity API, SAM3, PaddleOCR-VL on RunPod) and fine-tuned Llama 3.1 (8B/70B) with LoRA/QLoRA.
- Delivered AI solutions across Legal, Education, Recruitment, POS, Healthcare, Crypto, Social Media and Content Generation.

SELECTED PROJECTS

MekongAI Nexus – open-source multi-agent platform

github.com/pvbang/nexus

Python 3.12 · FastAPI · React 19 · TypeScript · PostgreSQL + pgvector · Redis · LiteLLM · Docker

- **Orchestration:** Planning → Think → Discuss → Execute → Report pipeline; agents run structured debate (challenge, rebuttal, synthesis), a beam-search tree planner with LLM-as-judge scoring, goal decomposition and verification, self-reflection checkpoints and a stuck detector.
- **Memory & RAG:** 4-layer memory (capture, atoms, scenes, persona) with a sleep-style consolidation engine; retrieval fuses vector, full-text and graph results with RRF (k=60) plus a cross-encoder reranker that degrades gracefully; experience-based continual learning (XSkill).
- **MCP:** client and JSON-RPC server with OAuth 2.1 + PKCE; 197 tool registrations across 66 modules in 29 toolsets, OpenAI-compatible API, 9 chat channels (Telegram, Discord, Slack, WhatsApp, Email, ...).
- **Reliability & security:** circuit breaker + provider fallback chain across 10+ LLM providers, task checkpoint and auto-resume, SSRF and exfiltration guards, prompt-injection data marking, Redis rate limiting, authorization gate for offensive-security tools.

- **Scale:** ~150K lines Python + ~64K lines TypeScript (React 19, 20 routes), 25 REST endpoint modules, 19 Alembic migrations, 12-service Docker Compose, 219 test files / 1,373 test functions.

AI Vehicle-Repair Assistant (client project)

LangGraph | FastAPI | MongoDB | Qdrant | Redis | AWS S3 | Streamlit | Docker

- LangGraph agent with 6 nodes (reasoning, search, evaluation, rerank, answer, verification) and conditional loops: 16-intent classifier, multi-DTC extraction, a coverage check that re-searches, and a verifier that retries the answer up to twice.
- Hybrid retrieval in Qdrant (1536-d dense + server-side IDF BM25 sparse vectors) with an LLM reranker that fills gaps via a tool loop; inline citations only for numbered references.
- 12 vehicle-data categories (specs, procedures, TSBs, wiring, labor times, ...) plus a 282-case repair-experience corpus; a separate bulk-LLM lane for a 250K-item background queue keeps chat latency and cost isolated.
- SSE streaming of every pipeline step, 264 API routes, per-user token and cost ledger, Streamlit admin UI; MongoDB backup to S3 with SHA-256 manifest and staging-first restore. 105K lines of source, 398 test files / 3,320 test functions.

M-Auto – AI YouTube video studio

Python | Streamlit | FFmpeg | faster-whisper | Edge-TTS | YouTube Data API | MCP

- 10-stage pipeline (topic, script, scenes, images, audio, video, metadata, quality, publish) with dependency-aware invalidation; LLM client with provider and model fallback and truncated-JSON repair.
- **AI quality critic** scores script, visual diversity, A/V sync and captions, then re-runs the weakest stage; deterministic guardrails for claims, language policy and clip duration.
- FFmpeg render engine (sub-pixel Ken Burns, xfade chains, karaoke captions from word-level Whisper timestamps, BGM ducking, 4 aspect ratios), DepthFlow 2.5D parallax, YouTube resumable upload with crash recovery, MCP bridge so agents can drive the editor. 49K lines Python, 825 test functions.

TikVault – AI video analysis and script generation

- Uploads or downloads (yt-dlp) TikTok/YouTube videos, sends them to Gemini and parses a Pydantic-validated JSON report: per-segment script breakdown, hook, pacing, technical audit and a 1–10 virality score; a generator then writes new scripts from analysed references with a creativity-to-temperature control.
- FastAPI + MongoDB backend with a range-request video proxy; React 19, Vite 7, Tailwind 4 and Recharts dashboard.

Customer-service RAG chatbot and game-lore assistant

- **Multi-tenant chatbot:** per-tenant Chroma stores, 7-class intent detection and entity extraction chained into routed prompts, order capture into Firestore, Messenger webhook with quick replies.
- **Eternal Odyssey assistant:** upload pipeline for 14+ file types (unstructured loaders, Vietnamese OCR), Qdrant retrieval and GPT-4o answers streamed over SSE, with spoiler-aware persona prompts.

Open-source developer tools

github.com/pvbang

- mcp-prune: strips cosmetic fields from MCP tool schemas, cutting tokens by 73.8% on the bundled fixture while validating the same inputs.
- sentinel-scan: zero-dependency static scanner for JS/TS (hardcoded secrets, eval/exec sinks, weak crypto).

MORE AI PROJECTS

- **AI Recruitment Platform**: CV parsing with OCR and a dual spaCy pipeline (vi + en); job-candidate matching by vector similarity (FastAPI, Qdrant, AWS S3).
- **AI Content Generator**: CrewAI pipeline (research, write, edit, publish) with RAG-based brand-voice learning for Facebook, Instagram and TikTok.
- **ESP32 AI voice bot**: hardware voice assistant controlling IoT devices through MCP and MQTT, with multi-LLM routing.
- **Capture2Text / Chrome extension**: capture on-screen text, then summarize, translate or answer with an LLM. **Vocab Master** and **TransVoiceHub** (React Native): LLM-powered English learning, speech-to-text, translation and TTS.

EDUCATION & LANGUAGES

B.Eng. in Information Technology · Vietnam-Korea University of ICT (VKU), Da Nang — 2020 – 2025

- Languages: Vietnamese (native), English (professional working proficiency: technical reading and writing).